RETRO COMPUTER FESTIVAL 2026

LUNA STATION

A home-made retro console on a tiny FPGA — featuring Lunar Lander



Luna Station is a hobby project celebrating the **60th anniversary of Luna 9's Moon landing**, created for the Retro Computer Festival in Cambridge.

FIRST GAME

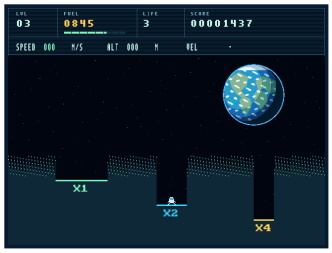
Lunar Lander is the console's first game. Rotate your spacecraft, fire the engine and land gently before the fuel runs out. Narrow pads score **×2** or **×4**, but land too fast or at an angle and you'll crash.

WHAT'S INSIDE?

An **FPGA** is a chip whose circuits can be configured to make a custom computer. Inside Luna Station are a tiny processor, memory, a video engine and an SD-card reader — all described in hardware logic.

> The game runs with just **8 KB of RAM** and uses integer "fixed-point" maths for its flight physics.





Real screenshots: main menu and a flight over the landing pads (640×480 HDMI).

TECH SPECS

BOARD	Sipeed Tang Nano 9K FPGA
CPU	RISC-V PicoRV32 · 27 MHz
VIDEO	HDMI 640×480 at 60 Hz
GAMES	FAT SD card · USB updates
SOUND	ESP32 · MAX98357A amplifier
CONTROL	ESP-NOW wireless gamepads

MOON FIRSTS

Luna 9 · 3 February 1966: the first automatic soft landing of a spacecraft on the Moon.

Apollo 11 · 1969: Neil Armstrong and Buzz Aldrin became the first people to walk on the Moon.

SCAN FOR THE SOURCE CODE, SCHEMATICS & 3D FILES

github.com/melrouts/RetroConsole-LunaStation

Eduard Garanskij · retro@lea.lt
Centre for Computing History, Cambridge

W START / THRUST
A D ROTATE
B MENU